

IT Relocation Site Survey, Service Agreement, Commercial RFQ, and Network Topology Blueprint

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Target Environment: NVIDIA Regional Hub — Brussels, Belgium

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Execution Runway: Strict 14 Calendar Days

Maximum Approved Budget Ceiling: €300,000.00 EUR

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1. IT Company Relocation Site Survey Plan & Physical Layout

Project Goal: To ensure the new building's infrastructure adequately supports the IT company's operational needs prior to relocation, minimizing downtime and facilitating a smooth transition.

1.1 Engineering Project Ownership

- **Madumathi Singaraju:** Lead Infrastructure Architect, Cyber Security Design Lead, & Technical Project Manager

1.2 Operational Survey Scope Analysis & Physical Space Audit

The comprehensive on-site facility audit mapped physical space constraints and media path requirements to establish full hardware deployment readiness:

- **Main Wiring Closet:** The central star distribution hub housing the Core Multilayer Switch, the edge firewall server stack, master patch panels, and primary server rack infrastructure.
- **IT Office (VLAN 60):** Configured with horizontal lines back to the core wiring closet for administrative hardware management and network orchestration.
- **Production Room (VLAN 10):** Outfitted with high-density cabling drops to run resource-heavy development and training computational tasks.
- **Support A (VLAN 40) & Support B (VLAN 20) Rooms:** Office spaces containing discrete access switches to manage client endpoint distribution lines. Terminated missing straight-through copper cabling to re-baseline disconnected workstations (PC6-PC9) and local shared printers.
- **Study Room (VLAN 30) & Management Office (VLAN 50):** User processing segments isolated into discrete logical VLAN spaces to optimize broadcast domains and limit network congestion.

2. Service Agreement Contract for IT Relocation and Infrastructure Upgrade

Effective as of June 2026, this binding agreement governs the project terms between the Service Provider (Madumathi Singaraju) and the client (Brussels Realty Group / NVIDIA regional steerage panels).

2.1 Scope of Services Summary

All infrastructure modifications, horizontal twisted-pair medium laydown, logical address subnetting, and service configurations are executed within an accelerated **14-calendar-day track** running continuous rolling shifts. The procurement process is managed natively, sourcing components from regional wholesale providers via **Dustin Belgium**.

2.2 Maintenance SLA and Liability

A **5-Year Continuous Maintenance and Technical SLA Package** is bundled into the delivery, covering remote auditing, system status reporting, automated remote off-site backups, and priority care for 60 months. Governing law rests strictly in accordance with the laws of Belgium (Brussels Region).

3. Quotation for IT Infrastructure Upgrade and Relocation Services

This section outlines the commercial quotation and technical cost estimations, optimizing base components via pre-negotiated wholesale enterprise discount metrics matching your network design parameters.

3.1 Itemized Capital Expenditure (CapEx) — Sourced via Dustin Belgium

Hardware / Sourcing Component	Description & Sourcing Blueprint Metrics	Qty	Total Price (€)
High-End Dell Workstations	Dell Computing Stack Fleet (Precision 3660/3460 & OptiPlex)	48	104,950.00
Core & Edge Networking Gear	1x Edge Firewall, 1x Cisco L3 MLS, 7x Cisco L2 Switches, Enclosed Racks, Smart UPS	10	36,400.00
Enterprise Software & Security Licenses	5x Windows Server Standard, 3-Year Unified UTM Security Suite, PRTG Monitoring	1	14,040.00
Structured Cabling Physical Media	66x Copper Straight-Through Lines (Cat6a Bulk Wire, Patch Panels, Wall Trays)	1	15,006.00
Hardened Peripheral Fleet	Network-Connected Enterprise Multi-Function Printers (MFPs)	4	18,000.00
TOTAL HARDWARE MATERIALS & EQUIPMENT BASE COST			188,396.00

3.2 Specialized Engineering Labor Breakdown (OpEx Taskforce Mapping)

Professional Engineering Resource Role	Estimated Project Hours	Total Cost (€)
IT Manager	80 Hours	12,800.00
Network Administrator	120 Hours	15,600.00
Lead Network Technician	160 Hours	16,000.00
Cyber Security Manager	45 Hours	7,654.00
Lead Security Technician	50 Hours	5,250.00
Lead RADIUS/AAA Configuration Specialist	40 Hours	6,000.00
TOTAL STAGING & TURNKEY LABOR VALUE		63,304.00

3.3 Financial Project Reconciliation Summary

- Subtotal Sourcing CapEx Investment: €188,396.00 EUR
- Subtotal Turnkey Labor Allocation: €63,304.00 EUR
- 5-Year Operational SLA Support Package (OpEx): €35,800.00 EUR

- **TOTAL ESTIMATED PROJECT PRICE TO CLIENT: €287,500.00 EUR** (€12,500.00 below the max €300k limit)
- **NET GROSS profit margin for our firm: €73,700.00 EUR (25.63%)**

4. Networking Logical Topology & Detailed Infrastructure

4.1 Logical Subnet Matrix & Variable Length Allocations

The network utilizes Class C private spaces engineered around an integrated Variable Length Subnet Masking (VLSM) strategy to map out corporate departmental spaces. All routing and security filters are processed dynamically at the core switch layer.

Subnet Zone Designation	VLAN ID	Network Address	Subnet Mask Size	Default Gateway IP	Central Core DNS Server
Production Sector	VLAN 10	192.168.10.0	255.255.255.0 (/24)	192.168.10.1	192.168.70.10
Support_B	VLAN 20	192.168.20.0	255.255.255.224 (/27)	192.168.20.1	192.168.70.10
Study Sector	VLAN 30	192.168.30.0	255.255.255.224 (/27)	192.168.30.1	192.168.70.10
Support_A Workstations	VLAN 40	192.168.40.0	255.255.255.224 (/27)	192.168.40.1	192.168.70.10
Management	VLAN 50	192.168.50.0	255.255.255.224 (/27)	192.168.50.1	192.168.70.10
IT Department	VLAN 60	192.168.60.0	255.255.255.240 (/28)	192.168.60.1	192.168.70.10
Server Pool & Storage SAN	VLAN 70	192.168.70.0	255.255.255.240 (/28)	192.168.70.1	Static (.10 Server Base)
Perimeter DMZ Zone	DMZ Interface	192.168.90.0	255.255.255.240 (/28)	192.168.90.1	192.168.70.10

4.2 Core Multi-Layer Switch Configuration (NVIDIA_SECURITY_ACL)

The Core Switch Virtual Interfaces (SVIs) process inter-VLAN routing, and automated dynamic identity management helper configurations. Access switch interface ports (such as `FastEthernet0/12`) are mapped as access links into their corresponding departmental VLAN matrices.

```
! Configure Core Inter-VLAN Gateway Routing SVIs
interface Vlan10
  ip address 192.168.10.1 255.255.255.0
interface Vlan20
  ip address 192.168.20.1 255.255.255.224
interface Vlan30
  ip address 192.168.30.1 255.255.255.224
interface Vlan40
  ip address 192.168.40.1 255.255.255.224
interface Vlan50
```

```
ip address 192.168.50.1 255.255.255.224
interface Vlan60
ip address 192.168.60.1 255.255.255.240
interface Vlan70
ip address 192.168.70.1 255.255.255.240
exit

! Enforce Extended Security Matrix to protect the iSCSI block storage target
ip access-list extended NVIDIA_SECURITY_ACL
10 permit ip 192.168.60.0 0.0.0.15 host 192.168.70.6
20 deny ip 192.168.10.0 0.0.0.255 host 192.168.70.6
30 deny ip 192.168.20.0 0.0.0.31 host 192.168.70.6
40 deny ip 192.168.30.0 0.0.0.31 host 192.168.70.6
50 deny ip 192.168.40.0 0.0.0.31 host 192.168.70.6
60 deny ip 192.168.50.0 0.0.0.31 host 192.168.70.6
70 permit ip any any
exit

interface Vlan70
ip access-group NVIDIA_SECURITY_ACL out
exit
```

4.3 Perimeter Firewall & DMZ Stateful Troubleshooting Ledger

To support corporate data assets, servers are assigned to specific functional blocks: the primary DNS Server runs statically at `192.168.70.10`, the centralized AAA/RADIUS authentication engine handles security management, and the high-throughput iSCSI block storage backend is locked to `192.168.70.6` under the security list restrictions. The public-facing web infrastructure is separated into a stateful perimeter DMZ zone.

A. Cisco ASA Firewall Perimeter Rule Tuning

To bypass asymmetric routing boundaries, internal subnets are summarized back to the core next-hop gateway. Concurrently, dynamic port exceptions are added to support the DMZ FTP Storage Server (`192.168.90.3`) and the corporate landing page Web Server (`192.168.90.2`).

```
! Inject summary backward path mapping to clear asymmetric drops
route inside 192.168.0.0 255.255.0.0 192.168.80.1

! Activate Stateful Packet Inspection for ICMP and FTP application layers
policy-map global_policy
  class inspection_default
    inspect icmp
    inspect ftp
exit
```

B. Post-Incident Diagnostic Verification Ledger

Diagnostic Test Vector	Action Taken	Expected Engineering Result	Actual Status Result
DNS Name Resolution	From client PC terminal, run: nslookup stratum.nvidia	Queries DNS host `192.168.70.10` and returns target `.90.2`	SUCCESSFUL — Forward lookup zone maps cleanly.
Web Landing Page Access	Open web browser and navigate to domain: stratum.nvidia	HTTP response renders the custom branded landing page portal	SUCCESSFUL — Cross-zone SVI and firewall traversal verified.
Stateful DMZ FTP Connection	From IT workstation terminal, run: ftp 192.168.90.3	Prompts for netadmin credentials via inspected port 21	SUCCESSFUL — Active login session established.
iSCSI Data Isolation Block	From standard Production PC, run: ping 192.168.70.6	Packet drops immediately via switch fabric filter	SUCCESSFUL — NVIDIA_SECURITY_ACL fully blocks lateral paths.

This master analysis report represents the definitive baseline engineering documentation for the NVIDIA Brussels Regional Hub network integration project. All milestones are 100% verified, compiled, and closed solely by Madumathi Singaraju.